

Co-Formation Games — Unifying Network Formation and Opinion Dynamics

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Abstract—This report proposes a novel model combining network creation studies and opinion dynamics. After introducing each Network Formation Games (NFGs) and Social Influence Models (SIMs), I compose a model allowing for co-evolution of opinions and networks and containing both NFGs and SIMs as special cases. Other sections consider basic dynamics, optimal states, and research directions of these Co-Formation Games (CFGs), as well as showcase an Julia module implementing CFGgames. Overall, the model offers a promising basis for future research projects, while broadening the perspective of NFGs and SIMs.

Index Terms—Collective dynamics, Game theory, Network formation games, Network theory (graphs)

I. INTRODUCTION

Network dynamics plays an ever-important role in describing the modern world, with connective plasticity and local optimizations enabling the emergence of global order. Since this framework can describe a vast variety of phenomena, researchers employed numerous approaches — often multi-disciplinary ones — to analyze these systems. Specifically, networks of human-like agents have been subject of studies ranging from mostly theoretical analysis to phenomenological social science works [1], [2].

Even as the sheer numerousness has inspired some authors to question each model's marginal benefit, considering strangs of research showcases a lack of holistic perspectives [1]: As a subfield of game theory, Network Formation Games (NFGs) study how self-interested agents create and drive the structure of networks [3]. In contrast, Social Influence Models (SIMs) focus on inner states of nodes (opinions) diffusing through it. In the case of social networks, this raises a tension: As opinion propagation and network formation can both be dynamic and influence each other, results from studies regarding only NFG or SIM do not necessarily generalize to the case of simultaneous dynamics.

This is an important caveat, as real social network behaviour rarely reduces to the semi-static treatment implied by NFGs and SIMs. Yet, even though there are works trying to combine some aspects of the other's influence, they often do so as a small part of the original model's main mechanism (e.g., [4], [5], [6]). Moreover, there is still no work unifying both approaches in a way sheltering results from both disciplines.

This report explores a model which allows topology and opinions to influence each other, while including NFGs and SIMs as special cases. After presenting NFGs and SIMs, this work presents and justifies each building block of the model, before assessing some basic traits with regard to possible dynamics, optimality and a simulation module implemented in the Julia programming language.

II. BACKGROUND

A. Network Formation Games

- Consider personal utility-maximising agents
- Action: Establish or cut edge sponsorship
- All actions (action profile) lead to Graph $G(a)$, hooray.
- Costs: Generally speaking, agents incur costs when sponsoring, and profit from closeness.
 - Overall cost function design like this...
- Stability concepts:
 - Nash equilibrium: No agent can increase utility via a single step. Think local equilibrium.
 - Optimal system: Graphs which minimise global costs.
 - PoA...
- Varieties:
 - Laterality – Directedness
- In terms of overlapping research:
 - Martin went for shapes opinions or sth. That's it.

B. Social Influence Models

(- Sidenote: Often used interchangeably with opinion dynamics, sorry Michael)

- Basic idea: Agents have opinions. Interaction = opinion exchange => Consequence
- Historically, most models are random interactions, i.e., have no topology.
- Some simple models
 - deGroot: Averaging real numbers – Ising model: we are all discrete spins – Modern stuff: Homophily, Similarity. — Structure implied, but not specific.
 - Bounded confidence et al. Because convergence was boring.
- Criticisms: No model captures true human behaviour so far. Duh.
- For our purpose: Edging closer to holistic model. No wish to capture human spirit or anything.

- Rather, see “opinion” as label for local information connected to edge; “social” as nodes influence each other due to “personal” information.

III. THE MODEL — CO-FORMATION GAMES

- This section shows the choices. Considered alternatives are in the appendix.
- Unilateral, Undirected
- TBW
-

A. Definitions & Equations

TABLE I
LIST OF VARIABLES USED IN THIS DOCUMENT

Variable	Symbol	Domain
Number of nodes	n	$\mathbb{N}$
Node of concern	i	$\{1, \dots, n\}$
Set of i 's neighbours	K_i	$\mathbb{N}^{ K_i }$
Network graph ¹	G	$\{0, 1\}^{n \times n}$
Opinion	x_i	$\mathbb{R}$
Opinion profile	x	$\mathbb{R}^{n \times n}$
Action	a_i	$\{0, 1\}^n$
Action profile	a	$\{0, 1\}^{n \times n}$
Costs per edge	α_c	$[0, \infty[$
Pairwise distance	d_{ij}	$\mathbb{N}$
Network valuation	v	$] - \infty, 0]$
Network expenses	f^c	$] - \infty, 0[$
Cost function	c	$] - \infty, 0[$
Communication rate	r_c	$[0, \infty[$
Formation rate	r_f	$\{0, 1\}$
Convergence threshold	ε	$[0, \alpha_c]$

where a norm $\|\cdot\|$ is the L0 norm, returning the count of non-zero elements.

a) Equations:

- Cost function

$$c_i = v_i(a) + f^c(a) \quad (1)$$

- Valuation function

$$v_i(a) = - \sum_{j=1, \dots, n} d_{ij} \quad (2)$$

- Network expenses:

$$f_i^c(a) = -\alpha_c \|a_i\| - \frac{1}{\|K_i\|} \sum_{j \in K_i} |x_i - x_j| \quad (3)$$

b) Algorithm:

TABLE II
THE PLANETS OF THE SOLAR SYSTEM AND THEIR AVERAGE DISTANCE FROM THE SUN

Planet	Distance (million km)
Mercury	57.9
Venus	108.2
Earth	149.6
Mars	227.9
Jupiter	778.6
Saturn	1,433.5
Uranus	2,872.5
Neptune	4,495.1

```

1 Initialise  $t = 1, c$ 
2 while  $(t < t_{\max} \wedge \varepsilon < \Delta c)$ 
3   if  $\text{mod}(t, r_c) = 0$  then
4     select random node  $i$ 
5      $x_i$  arrow.l get opinion update  $x_i^u$ 
6   end if
7   if  $r_f = 1$  then
8     select random node  $i$ 
9      $a_i$  arrow.l get action update  $a_i^u$ 
10  end if
11   $t += 1$ 
12   $c(t)$  arrow.l get new costs  $c_t$ 
13   $\Delta c = c_{t-1} - c(t)$ 
14 end while

```

Algorithm 1. Dynamics of our synchronous move game

IV. OLD STUFF FROM THE TEMPLATE



Fig. 1. A circle representing the Sun.

In Fig. 1 you can see a common representation of the Sun, which is a star that is located at the center of the solar system.

In Table II, you see the planets of the solar system and their average distance from the Sun. The distances were calculated with (1) that we presented in

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¹The graph is represented as an adjacency matrix.

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